

IP Practical Exam

Code:

server.js

```
const express = require('express');
const bodyParser = require('body-parser');
const cors = require('cors');

const app = express();
const PORT = 5000;

app.use(cors());
app.use(bodyParser.json());

app.post('/calculate-interest', (req, res) => {
  const { principal, age, investmentPeriod } = req.body;

  const interestRate=10;

  const interestEaarned = principal * interestRate * investmentPeriod;
  const finalage=age + investmentPeriod;
  res.json({ interestEaarned });
  res.json({ finalage });

});

app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

App.js

```
import React, { useState } from 'react';
import './App.css'
const App = () => {
  const [principal, setPrincipal] = useState("");
  const [age, setAge] = useState("");
  const [investmentPeriod, setInvestmentPeriod] = useState("");
  const [interestEaarned, setInterestEarned] = useState(0);

  const handleSubmit = async (e) => {
    e.preventDefault();

    const response = await fetch('http://localhost:5000/calculate-interest', {
      method: 'POST',
      headers: {
        'Content-Type': 'application/json',
      },
      body: JSON.stringify({
        principal: parseFloat(principal),
        age: parseInt(age),
        investmentPeriod: parseInt(investmentPeriod),
      }),
    });

    const data = await response.json();
    setInterestEarned(data.interestEaarned);

  };

  return (
    <div className='main'>
      <h1>Savings Interest Calculator</h1>
```

```
<form className='forminp' onSubmit={handleSubmit}>
  <input className='inp'
    type="number"
    placeholder="Principal Amount"
    value={principal}
    onChange={(e) => setPrincipal(e.target.value)}
    required
  />
  <input className='inp'
    type="number"
    placeholder="Age"
    value={age}
    onChange={(e) => setAge(e.target.value)}
    required
  />
  <input className='inp'
    type="number"
    placeholder="Investment Period (years)"
    value={investmentPeriod}
    onChange={(e) => setInvestmentPeriod(e.target.value)}
    required
  />
  <button className='inp button ' type="submit">Calculate Interest</button>
</form>
```

```
    {interestEarned > 0 && <h2>Interest you will get
    :₹{interestEarned.toFixed(2)} when you will turn
    {parseInt(age)+parseInt(investmentPeriod)}</h2>}
    </div>
  );
};
```

```
export default App;
```

Name:Atharv Nlkam

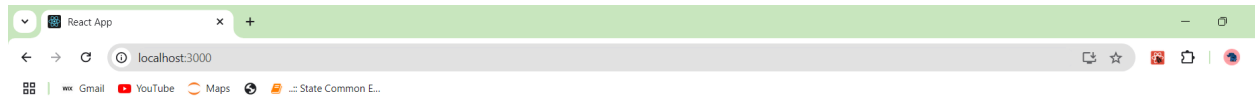
Div:D15C

Roll:36

OUTPUT:

Savings Interest Calculator

Principal Amount	Age	Investment Period (years)	Calculate Interest
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Savings Interest Calculator

102	2	12	Calculate Interest
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Interest you will get :₹12240.00 when you will turn 14